

A Geometric Knowledge Oriented Single-Frame 2D-to-3D Human Absolute Pose Estimation Method

Mengxian Hu^{1b}, Chengju Liu^{1b}, Shu Li^{1b}, Qingqing Yan^{1b}, *Graduate Student Member, IEEE*, Qin Fang,
and Qijun Chen^{1b}, *Senior Member, IEEE*

Abstract—As a critical part of the 3D human pose estimation (HPE), establishing the 2D-to-3D lifting mapping is limited by depth ambiguity. Most current works generally lack the quantitative analysis of the relative depth expression and the depth ambiguity error expression in lifting mapping, resulting in low prediction efficiency and poor interpretability. To this end, this paper mines and leverages prior geometric knowledge of these expressions based on the pinhole imaging principle, decoupling the 2D-to-3D lifting mapping and simplifying the model training. Specifically, this paper proposes a prior geometric knowledge oriented pose estimation model with two-branch transformer architectures, explicitly introducing high-dimensional prior geometric features to improve model efficiency and interpretability. It converts the regression of spatial coordinates into the prediction of spatial direction vectors between joints to generate multiple feasible solutions further alleviate the depth ambiguity. Moreover, this paper raises a novel non-learning-based absolute depth estimation algorithm based on prior geometric relationship decoupling from relative depth expression for the first time. It establishes multiple independent depth mapping from non-root nodes to the root node to calculate the absolute depth candidate, which is parameter-free, plug-and-play, and interpretable. Experiments show that the proposed pose estimation model achieves state-of-the-art performance on Human 3.6M and MPI-INF-3DHP benchmarks with lower parameters and faster inference speed, and the proposed absolute depth estimation algorithm achieves similar performance to traditional methods without any network parameters. The source code are available at <https://github.com/Humengxian/GKONet>.

Index Terms—Pose estimation, absolute depth, pinhole imaging, geometric feature, transformer.

I. INTRODUCTION

ESTIMATING the 3D human pose from a monocular image is a fundamental vision task with broad applications, such as action recognition, virtual reality, and motion prediction [1], [2], [3]. The typical pose estimation method is detecting a series of 2D key points from the original image and then lifting these 2D key points to 3D key points [4], [5], [6]. Still, its accuracy is limited by an ill-posed problem due to the ambiguity of the depth value in 2D joint representation.

Expanding temporal information and decoupling spatial features are two aspects to alleviate the abovementioned problem [7], [8], [9]. For example, MHFormer [9] learned spatial-temporal hypotheses by different attention modules, and P-STMO [10] utilized the pre-training method to obtain spatial-temporal dependence. These video-based methods generally employ a temporal network to establish the spatial-temporal consistency of the human pose, thus reducing the error caused by depth ambiguity. However, there are two limitations. First, the temporal model expands the model size and computation burden, leading to real-time challenges. The second and most crucial point is that its preconditions are relatively harsh in applications. Common video-based methods generally require 81 or 243 consecutive frames of video, and real-time detection of 2D key points in these consecutive frames requires numerous computing resources. In addition, some time-sensitive tasks, such as fall detection, require instantaneous human pose estimation. Such scenes have raised the demand to estimate 3D human poses from a single frame image.

Most existing single-frame methods tend to design particular network architecture for learning the high-dimensional spatial geometric features. They either employ the graph convolution network [11], [12] to build the graph architecture representation of the human skeleton or apply the transformer-based model [13], [14] to construct the connection between different joints adaptively. Although some well-designed network architectures have been proven effective with adequate parameters, they lack the directly utilize of prior geometric knowledge and neglect the quantitative analysis of depth ambiguity error. Thus, numerous network parameters are paid for mining high-dimensional geometric knowledge and singularity causes, reducing

Manuscript received 27 April 2023; accepted 16 May 2023. Date of publication 23 May 2023; date of current version 7 December 2023. This work was supported in part by the National Natural Science Foundation of China under Grant 62173248, Grant 62073245, and Grant 62233013; in part by the Suzhou Key Industry Technological Innovation-Core Technology Research and Development Program under Grant SGC2021035; in part by the Special Funds for Jiangsu Science and Technology Plan under Grant BE2022119; in part by the Shanghai Municipal Science and Technology Major Project under Grant 2021SHZDZX0100; and in part by the Fundamental Research Funds for the Central Universities, Shanghai Science and Technology Innovation Action Plan, under Grant 22511104900. This article was recommended by Associate Editor X. Guo. (Corresponding authors: Chengju Liu; Qijun Chen.)

Mengxian Hu is with the Shanghai Research Institute for Intelligent Autonomous Systems, Tongji University, Shanghai 201804, China, and also with the Robotics and Artificial Intelligence Laboratory (RAIL), Tongji University, Shanghai 201804, China (e-mail: humengxian@tongji.edu.cn).

Chengju Liu is with the Robotics and Artificial Intelligence Laboratory (RAIL), Tongji University, Shanghai 201804, China, and also with the Tongji Research Institute of Artificial Intelligence (Suzhou), Suzhou 215300, China (e-mail: liuchengju@tongji.edu.cn).

Shu Li, Qingqing Yan, Qin Fang, and Qijun Chen are with the Robotics and Artificial Intelligence Laboratory (RAIL), Tongji University, Shanghai 201804, China (e-mail: lishu@tongji.edu.cn; qyan_0131@tongji.edu.cn; tongji_fq@tongji.edu.cn; qjchen@tongji.edu.cn).

Color versions of one or more figures in this article are available at <https://doi.org/10.1109/TCSVT.2023.3279291>.

Digital Object Identifier 10.1109/TCSVT.2023.3279291

inference speed and accuracy. Concurrently, additional parameters in learning-based absolute depth estimation are required to integrate the global depth features, further decreasing model real-time and interpretability.

Considering the above limitations, this paper mainly studies the mathematical expression of the relative depth between joints based on the inverse problem of the pinhole imaging model, quantitatively analyzing the causes and factors of singularity in the 2D-to-3D lifting mapping. The first conclusion is that the geometric features between nodes and internal camera parameters can explicitly represent the relative depth of the sub-node in the two-node pinhole imaging system, which is fully decoupled. And the depth singularity appears in the possibility that the triangle determined by the “*Side, Side, Angle(SSA)*” may have double solutions when solving the inverse problem. Moreover, the above geometric features, such as the projection coordinates and spatial included angle, directly affect the singularity error caused by depth ambiguity. Introducing these prior geometric features into the pose estimation model can effectively simplify the 2D-to-3D lifting mapping and comprehensively learn the singularity error laws, ensuring great accuracy with low parameters. The second conclusion is that given the relative coordinate between two nodes, projection coordinates, and internal camera parameters in the same above system, the absolute depth of any node can be solved in reverse. It means that deterministic mathematical expressions can complete absolute depth estimation of relative 3D poses without any network parameters. Compared with traditional methods, the non-learning-based method based on this discovery can improve the model’s efficiency and interpretability.

To this end, this paper proposes a novel model to estimate 3D human pose from a monocular image based on the first conclusion. The critical insight is to allow the model to learn the pre-extracted prior geometric features efficiently. To accomplish this, the model employs a two-branch transformer-based architecture, and each branch consists of several stacked modules comprising a feature fusion block and a transformer encoder. Such an architecture relies on the transformer’s powerful global modeling capabilities to extract projection coordinate features and prior geometric features, respectively, and integrate semantic features of another branch through its feature fusion modules. The model outputs the 3D vector pointing from each joint to the other joints and finally generates multiple feasible solutions to reduce further the error caused by depth ambiguity.

Next, this paper proposes a non-learning-based absolute depth estimation algorithm based on the second conclusion. Although the geometric relationship expression can calculate the absolute depth candidate values, these values are directly affected by the relative position error and depth ambiguity. To solve this problem, this paper utilizes the trimmed mean method to ensure robustness under different scenes. High-precision absolute-depth information can meet the demand of some industrial application scenarios and calibrate the existing relative 3D pose through the positive problem of pinhole imaging [15]. Since no additional network parameters are required, the algorithm can be extended to any existing

3D human pose estimation model with excellent application significance.

The experiment shows that the proposed pose estimation model achieves state-of-the-art precision under lower parameters and faster inferring speed on two popular benchmarks: Human3.6M and MPI-INF-3DHP. Furthermore, the proposed absolute depth estimation algorithm achieves similar performance to the learning-based method without any network parameters, which is perfectly adapted to existing single-frame and video-based pose estimation models. This paper’s contributions can be summarized as follows:

- This paper derives a fully decoupled and geometry-based expression of relative depth estimation in the pinhole imaging system, further quantitatively analyzing the cause and factors of depth ambiguity.
- A novel two-branch transformer-based model is proposed to estimate 3D human pose from a monocular single-frame image. It explicitly introduces prior geometric features and generates multiple feasible solutions, reducing the complexity of 2D-to-3D lifting mapping and the error of depth ambiguity.
- This paper presents, to the best of our knowledge, the first non-learning-based and geometric-based absolute depth estimation algorithm for 3D human pose estimation. It is parameter-free, plug-and-play, and interpretable.
- The proposed pose estimation model achieves state-of-the-art results with fewer parameters and faster inference speed on two 3D human pose estimation benchmarks.

II. RELATED WORK

Estimating 3D human poses from images is a long-standing computer vision task. Before the rise of deep learning, 3D human pose estimation tasks were mainly based on traditional machine learning methods such as the bayesian model [16], the support vector machine [17], etc. Subsequently, pose estimation methods based on deep learning have emerged, bringing colossal success [13], [18]. These deep learning methods are generally divided into two categories: one-stage methods and two-stage methods.

The one-stage methods [19], [20], [21] utilize of depth learning model to directly predict the coordinates of the 3D human joint from the image. Most works regarded the one-stage method as a regression task of the joint coordinates. Li et al. [19] first proposed the end-to-end convolutional network framework for 3D human pose estimation from single-frame images. Pavlakos et al. [22] established the voxel representation of 3D human posture to reduce the mapping complexity. Yang et al. [23] constructed a 3D human pose generator based on the adversarial learning framework, improving the model’s accuracy. Despite these efforts, mapping from images to 3D joint coordinates remains complex, challenging model accuracy and generalization capabilities.

The two-stage estimation method [11], [23], [24] detects the 2D human pose first and then lifts from the 2D pose to the 3D pose using different deep learning models. This method is compatible with the 2D pose estimation model, and it is helpful to apply data augmentation. Early work [5] employed

a naive multi-layer perceptron to extract joint information without additional design. Ghezelghieh et al. [25] proposed a convolution network framework to combine camera viewpoint and 2D joint locations. Subsequent works argued that the human skeleton should be regarded as a graph architecture. Graph convolutional layers are used as feature extraction modules to apply prior knowledge of human skeleton connections [6], [11], [12], [26]. Xu and Takano [11] constructed a graph network based on auto-encoder architecture to extract multi-scale features of the human skeleton. Zhu and Tang [12] introduced the weight modulation method to reduce the model size of the graph network. Zeng et al. [27] proposed a hop-aware hierarchical channel-squeezing fusion layer and temporal-aware dynamic graph module to provide targeted training for hard samples, achieving low redundancy and high accuracy. Other works have recently employed the multi-head self-attention module or MLP-Like architecture to establish joint interconnection [13], [14], [28]. Zheng et al. [28] first utilized the pure transformer architecture to solve the 2D-to-3D lifting task. Lutz et al. [13] established the error transformer and introduced the intermediate supervision training strategy to improve the model's accuracy. Li et al. [29], inspired by MLP Mixer [30], proposed a spatial graph MLP block and channel graph MLP blocks for pose estimation, which balance the skeletal representation of globality and locality. Some works, such as Pavllo et al. [31], noted the usefulness of pinhole imaging models in 2D-to-3D lifting tasks and designed special loss functions. Nevertheless, these efforts lack the direct use of geometric features in pinhole imaging principles. Moreover, the 2D-to-3D lifting mapping is an ill-posed problem. None of the above work studies the properties of the mapping itself mathematically. Especially there is an insufficiency of quantitative analysis of the singular errors in mapping. This paper embeds prior geometric features into the model and quantitatively discusses the causes of depth ambiguity in 2D-to-3D lifting mapping to address the above problem.

At present, only a few works [31], [32], [33] have specifically discussed the absolute coordinate prediction of the 3D pose. Pavllo et al. [31] used the trajectory model to predict the absolute coordinates of the root node. Zhan et al. [33] transformed the input space from two-dimensional pixel space to 3D rays in a standardized coordinate system and achieved robust prediction of absolute coordinates. The common point of these works is to learn the root node coordinate information from the global features by designing different network architectures, which leads to increased model parameters and decreased interpretability. This paper presents a novel method for calculating absolute depth based on geometric relations. It overcomes the dependence of the above traditional methods on additional parameters, improving the efficiency and interpretability of the model.

III. DEPTH REPRESENTATION IN PINHOLE IMAGING

Based on the pinhole imaging model, this section studies the explicit expression of relative depth between joints. Let $A = (X_1, Y_1, Z_1)$ and $B = (X_2, Y_2, Z_2)$ be two human joints connected by bones in three-dimensional space, and

their projection on the plane are $C = (u_1, v_1)$ and $D = (u_2, v_2)$ respectively. It is generally considered that the distance between them remains unchanged.

According to the pinhole imaging model, the mapping between plane pixels and spatial coordinates without considering distortion can be expressed as Equation 1.

$$Z_i \begin{bmatrix} u_i \\ v_i \\ 1 \end{bmatrix} = \begin{bmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} X_i \\ Y_i \\ Z_i \end{bmatrix} \quad (1)$$

where f_x and f_y are the focal length, and c_x and c_y are the principal point coordinates of the camera. For convenience, both c_x and c_y are equal to zero. When the model input is only the absolute coordinates of the joint on the plane, an obvious optimization method is to solve the following equations.

$$X_i = \frac{Z_i}{f_x} u_i \quad (2)$$

$$Y_i = \frac{Z_i}{f_y} v_i \quad (3)$$

$$AB^2 = (X_1 - X_2)^2 + (Y_1 - Y_2)^2 + (Z_1 - Z_2)^2 \quad (4)$$

Substitute Equation 2 and Equation 3 into Equation 4, and the relationship about depth Z_1 , Z_2 is highly coupled. Obviously, it is challenging to learn adequate relative depth information, which increases the difficulty of model training.

This paper finds another more straightforward optimization method based on geometric features. The visualization of the pinhole imaging model is shown in Figure 1. A local coordinate system at the joint A is established, and the plane Ψ (X_A - A - Y_A) is parallel to the imaging plane Φ (x - o - y). Set the intersection of the line BD and the plane Ψ as point E , the intersection of the Z -axis and the plane Ψ is point G , and the vertical foot of B joint in the plane Ψ is point F . Due to the excellent property of parallel planes, it can be deduced that triangular CDO is similar to triangular AEO and triangular CoO is similar to triangular AGO . It means that Equation 5 is established.

$$\frac{AE}{CD} = \frac{AO}{CO} = \frac{GO}{oO} = \frac{Z_1}{f} \quad (5)$$

According to the rules of triangle angles, Equation 6 can be obtained. To simplify the equation, this paper lets the $\angle AEB$ as α and the $\angle ODC$ as β .

$$\alpha = \begin{cases} \pi - \angle AEO = \pi - \beta, & \text{if } Z_2 \geq Z_1, \\ \beta, & \text{if } Z_2 < Z_1. \end{cases} \quad (6)$$

Then the length of BE can be calculated based on the cosine theorem.

$$BE^2 - 2AE \cdot \cos \alpha \cdot BE + AE^2 - AB^2 = 0 \quad (7)$$

$$BE = AE \cdot \cos \alpha \pm \sqrt{AE^2 \cdot \cos^2 \alpha - (AE^2 - AB^2)} \quad (8)$$

It can prove similarly that triangular oDO is similar to triangular FEB . Further, the explicit expression of the relationship between Z_1 and Z_2 can be expressed as Equation 9.

$$Z_2 = BF + Z_1 = f \cdot \frac{BE}{OD} + Z_1 \quad (9)$$

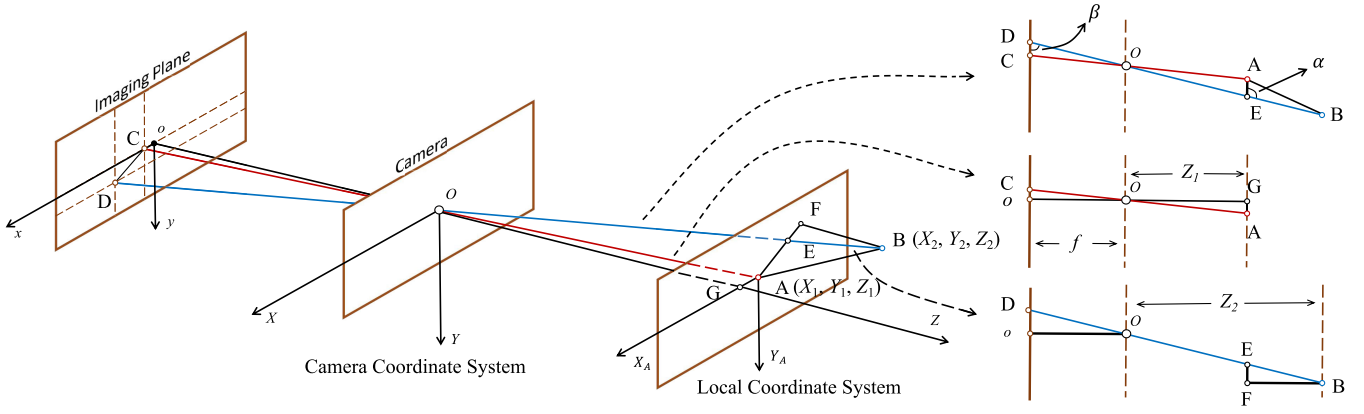


Fig. 1. The diagram of a two-node pinhole imaging math model. It describes any two points A and B in three-dimensional space projected as points C and D in the imaging plane through the aperture O . Establish a plane Ψ (X_A - A - Y_A) parallel to the imaging plane through point A . The extension of DO and z -axis intersect the plane Ψ at E and G , respectively. Point F is the vertical foot of B in the plane Ψ . The spatial location relationship of A and B , such as the distance of AB or the relative depth of B and A , can be simplified as the three sets of plane relationships shown in the right part of the diagram.

The depth information of joint B is not included on the right side of Equation 9, so it can be considered that the related depth information described in Equation 9 is wholly decoupled, that is, $Z_2 = h(Z_1)$. Simultaneously, it can be noticed that the positive and negative signs in Equation 8 will cause a singularity in the results. This singularity stems from the fact that the “Side, Side, Angle (SSA)” condition cannot form a unique triangle AEB under general conditions. Specifically, when α is an acute angle, and AB is less than AE , the solution of relative depth will appear as such error shown in follow Equation 10.

$$error = 2\sqrt{\left(\frac{Z_1 \cdot CD}{f}\right)^2 (\cos^2 \alpha - 1) + AB^2 \cdot \frac{f}{OD}} \quad (10)$$

This paper considers f as a constant. At this time, the projection distance CD , the depth Z_1 of the parent node, and the distance OD from the child node to the origin are inversely proportional to the error. Furthermore, the included angle α and the distance AB between nodes are directly proportional to the error. In 3D human pose estimation, the geometric features, such as CD , OD , and $\cos \alpha$, can be solved by the following equations.

$$CD = \sqrt{(u_1 - u_2)^2 + (v_1 - v_2)^2} \quad (11)$$

$$OD = \sqrt{u_2^2 + v_2^2 + f^2} \quad (12)$$

$$\cos \alpha = \pm \frac{-u_2(u_1 - u_2) - v_2(v_1 - v_2)}{\sqrt{u_2^2 + v_2^2 + f^2} \sqrt{(u_1 - u_2)^2 + (v_1 - v_2)^2}} \quad (13)$$

Refer to Equation 6 for the meaning of positive and negative signs in Equation 13. One of the main motivations of this paper is to efficiently utilize the prior geometric features in Equation 9 and Equation 10. A simple idea from feature engineering is to feed these well-chosen geometric features into the network, which guides the network actively and adaptively in searching for the relationship between the geometric features and the model output. The benefits of this design are twofold. One is that by incorporating known geometric characteristics, the complexity of the mapping can be reduced according to Equation 9, and the model can avoid wasting computation

in nonlinear relationships with explicit expressions. Secondly, it is easier for the model to learn the law of error when encountering depth singularity. These advantages ensure that the network performs better precision and generalization performance with lower parameters during pose estimation.

On the other hand, when the relative depth between joints is known, the absolute depth of the last joint can be calculated. Follow equations are established according to Equation 5, Equation 7, and Equation 9.

$$BE = OD \cdot \frac{|Z_2 - Z_1|}{f} \quad (14)$$

$$AE = BE \cdot \cos \alpha \pm \sqrt{BE^2 \cdot \cos^2 \alpha - (BE^2 - AB^2)} \quad (15)$$

$$Z_1 = f \cdot \frac{AE}{CD} \quad (16)$$

In this case, the symbol selection in Equation 15 still comes from the two possible solutions for the triangle AEB caused by the “Side, Side, Angle (SSA)” condition after determining BE , AB , and angle α . Specifically, depth ambiguity may occur when α is acute and AB is less than BE . When the symbol is determined, the absolute depth Z_1 of the parent node can be directly calculated from Equation 16, which is an exciting conclusion. From these equations, a non-learning-based algorithm is proposed in Section IV-B, which is efficient and highly interpretable.

IV. METHOD

A. Pose Estimation Model

The overview of the proposed model is depicted in Figure 2 (a). Given a single-frame 2D human pose, the model encodes the geometric features between its joints in pairs. Its two transformer branches are used to handle the projected coordinate and prior geometric features, respectively. Each fusion module in the backbone owns independent network parameters to ensure efficient feature integration. For convenience, we refer to the 3D vector in space starting from node A and ending at node B as the spatial direction vector between the node A and node B . The head network outputs the spatial direction vectors between every two joints in pairs to keep the

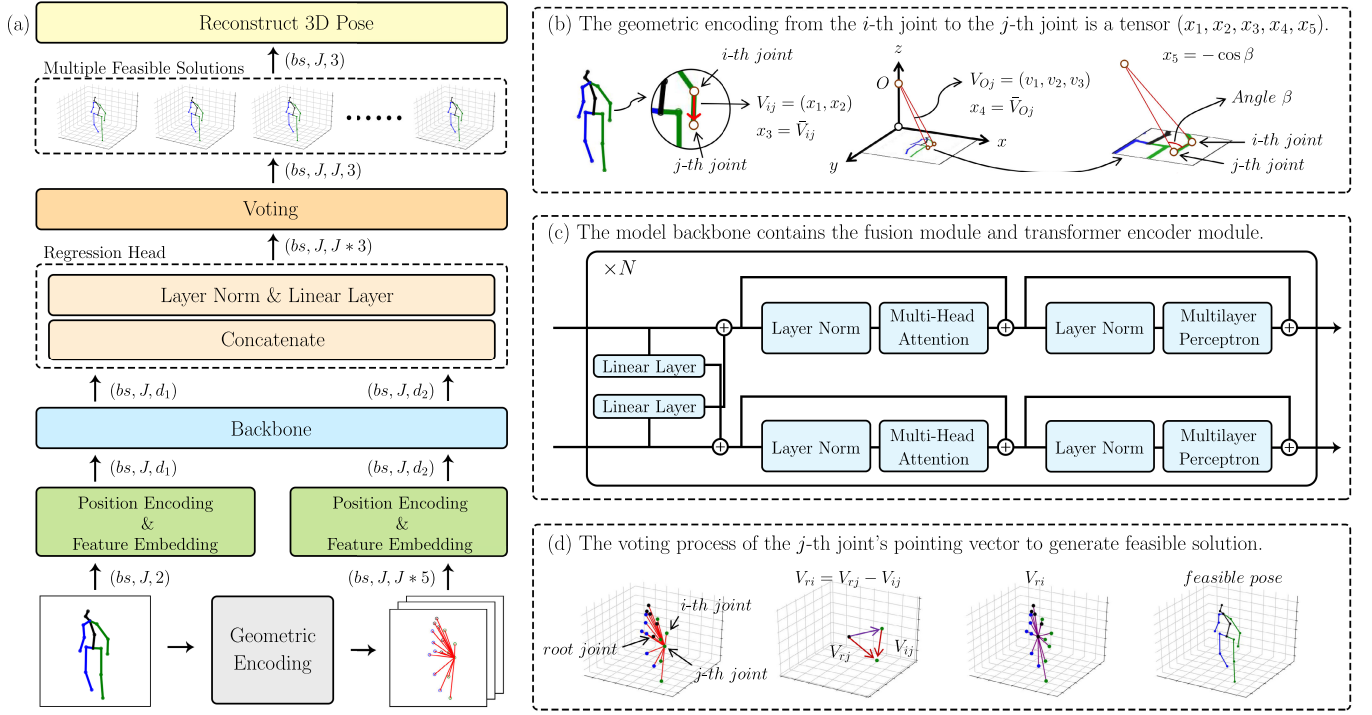


Fig. 2. (a) Overview of the proposed model. (b) An example of the geometric encoding tensor contains five dimensions. Our model requires encoding geometric information between any two joints to obtain as many features as possible. (c) Backbone architecture diagram of the proposed model. The two branches communicate through a linear layer and use two parallel transformer encoding modules to extract features. (d) A standard voting process for generating feasible solutions. All vectors pointing to any joint are converted into new vectors starting from the root node (usually hip).

semantics consistent with the geometric feature input. These vectors are used to generate multiple feasible solutions and the final 3D human pose. More model details and motivations are reported as follows:

1) *Default Configuration*: The proposed model's hyper-parameters include the number of joints J , the dimension d_1 of the projected coordinate branch (the first branch), the dimension d_2 of the prior geometric feature branch (the second branch), the number of layers N of the backbone, and the activation function σ of the multi-layer perceptron. In this paper, default parameters set J as 17, d_1 as 32, d_2 as 128, and N as 8. This paper employs a smaller d_1 to force the model to pay more attention to geometric features. Moreover, all activation functions of the multi-layer perceptron are set to the GELU function.

2) *Geometric Encoding*: According to the analysis in Section III, this paper mainly encodes the prior geometric features from the i -th joint to the j -th joint in five dimensions, such as Figure 2 (b). The first two dimensions are plane vector CD of the i -th joint pointing to the j -th joint, and the third dimension is its modulus. Such a plane vector can be intuitively understood as the projection of a spatial direction vector on an imaging plane, which preserves directional information perpendicular to the depth direction. Next, the fourth dimension is the modulus of the space vector DO of the j -th joint pointing to the camera aperture. Finally, the fifth dimension is the inverse of the cosine of the angle between the space vectors DC and DO . The last three dimensions are extracted from depth expression Equation 9 and error expression Equation 10. The pre-calculated initial

prior geometric feature tensor has dimensions $(batch\ size, J, J, 5)$, referring to the five-dimensional geometric features between any two nodes. This paper merges the third and fourth dimensions of the prior geometric feature tensor before feeding to the embedding layer. It ensures that the input dimensions of the second branch are consistent with those of the first branch.

3) *Backbone*: The proposed method employs a two-branch transformer-based module as the backbone. Each branch contains a series of stacked modules consisting of a feature fusion layer and a transformer encoder. The feature fusion layer is based on linear mapping and element-by-element addition. It effectively exchanges the feature of the two branches, giving the model a wider receptive field. The transformer encoder is referred from [34], which can dynamically extract sequence features and adaptively model human skeleton relates information. The model directly utilizes its four basic components without modification: position embedding, layer normalization, multi-head self-attention, and multi-layer perceptron.

4) *Regression Head*: Unlike the point-to-point 2D-to-3D lifting mapping established by traditional methods, the proposed model innovatively constructs the tensor lifting mapping between joints from prior geometric features to spatial direction vectors. Such mapping maintains the consistency of input and output semantics. It can be understood as deriving the relative position relationship between any two nodes from their geometric feature relationship. Such spatial direction vector has a clear geometric significance. It represents the candidate position of other nodes from the current node perspective. To this end, the regression head network concatenates the features extracted by the backbone and outputs the spatial

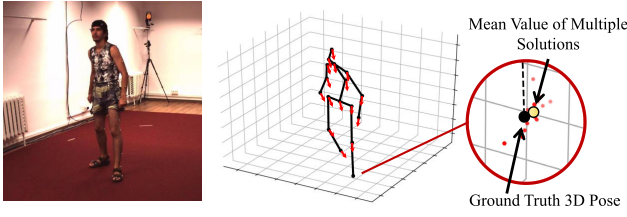


Fig. 3. Visualize the results of multiple feasible solutions and the final 3D pose generated by the proposed model. The red points denote the conjecture of each joint about the possible position of the left foot, the yellow point denotes the mean value of the feasible solution, and the black point denotes the ground truth coordinate.

direction vectors from any node to other nodes using a shared parameter linear layer.

5) *Voting*: The proposed model utilizes the spatial direction vectors to generate J sets of feasible solutions. These feasible solutions are similar to voting results and are inferences of the overall 3D pose from different points of view, as shown in Figure 3. In application, the ground truth pose of the human is unknown, but the root node is generally located at the origin. The proposed model generates hypothetical poses by converting all joint vectors into relative positions from the root node to other nodes. Figure 2 (d) shows a feasible solution generation case. The final 3D pose can be obtained by calculating the mean value of the coordinates of these feasible solutions at each joint.

6) *Loss Function*: The optimization objective of the proposed model is to minimize the error between the predicted spatial direction vectors V and the ground truth spatial direction vectors $\hat{V}$. Therefore, this paper designs the following loss function without bells and whistles.

$$loss = \sum_{i=1}^J \sum_{j=1}^J \|V_{ij} - \hat{V}_{ij}\|_2 \quad (17)$$

B. Absolute Depth Estimation Method

Current learning-based root node's absolute depth estimation methods [5], [12], [15] are usually considered a 3D human pose estimation sub-task. It shares parts of network parameters with the 3D relative pose estimation branch and has its own loss function. However, absolute depth estimation is usually more complex than relative pose estimation because the former has a broader solution space and lacks constraints of prior skeletal knowledge. Therefore, the traditional method requires more network parameters and carefully adjusts the hyperparameters to balance the two prediction tasks.

This paper presents a non-learning-based absolute depth estimation algorithm by decoupling the geometric relationship between the joints. In detail, it establishes multiple independent mapping from each non-root node's 3D pose, 2D pose, and the camera's internal parameter to the root node's absolute depth based on Equation 16. Theoretically, the depth candidate values calculated from each joint are limited by the symbol ambiguity in Equation 15 and the accuracy of 3D relative pose estimation. This paper adopts the trimmed mean method to alleviate the above limitations.

Algorithm 1 Absolute Depth Estimation of the Root Node

Input:

- J : the number of joints.
- k : the trimmed coefficient.
- f : the focal length of camera.
- $\{p_j = (u_j, v_j), j = 1, \dots, J\}$: the 2D pose.
- $\{P_j = (X_j, Y_j, Z_j), j = 1, \dots, J\}$: the 3D relative pose.

Output:

- z : the absolute depth of the root node.

Assume:

- $j = 1$ is the index of the root node.

```

1: for  $j \in \{2, \dots, J\}$  do
2:    $OD_j = \text{sqrt}(u_j^2 + v_j^2 + f^2)$ 
3:    $BE_j = OD_j \cdot Z_j / f$ 
4:    $CD_j = \text{sqrt}((u_j - u_1)^2 + (v_j - v_1)^2)$ 
5:    $Cos_j = (u_j \cdot (u_1 - u_j) + v_j \cdot (v_1 - v_j)) / (OD_j \cdot CD_j)$ 
6:    $AB_j = \text{sqrt}(x_j^2 + y_j^2 + z_j^2)$ 
7:    $AE_j = \text{sqrt}(BE_j^2 \cdot (Cos_j^2 - 1) + AB_j^2) + BE_j \cdot Cos_j$ 
8:    $z_j = f \cdot AE_j / CD_j$ 
9: end for
10:  $z_j = \text{sort}(z_j)$ 
11:  $z = \sum_{i=k}^{N-k} (z_i / (J - 2k))$ 

```

For symbol ambiguity in Equation 15, it has a probability of occurrence when α is acute, and AB is less than BE , according to the analysis in section III. Intuitively, the probability of choosing a negative sign in Equation 15 tends to be low. This paper provides statistical data from two real datasets in Section V-B to support the above viewpoint. The simple idea based on the trimmed mean skill is to utilize the positive sign in all the above mappings and then delete k maximum depth candidates. This way, even if symbol ambiguity exists and the positive sign is selected as an error, the algorithm can filter these error depth values in time.

For the inaccurate 3D relative pose, its relative depth $Z_2 - Z_1$ (in Equation 14) and distance AB (in Equation 15) from non-root nodes to root nodes will induce inaccurate depth candidate values, which usually occur in occlusion cases or blur cases. In these cases, the depth candidate values calculated from these inaccurate joints will deviate significantly from the actual depth values. The trimmed mean method removes both the maximum and minimum k values and then averages the remaining candidate values. It can filter out some inaccurate joint inference depth candidate values. Even if there are candidate values with significant errors in the remaining list of depth candidate values, their errors will be reduced by the averaging operator.

The root node absolute depth estimation can be summarized as Algorithm 1. The advantages of the proposed algorithm are parameter-free, plug-and-play, and interpretability. Its floating-point computation is less than 1000 times, far less than the matrix multiplication in the traditional method. Simultaneously, its input is not limited so that it can be easily extended to any existing 3D human pose estimation model. Finally,

it opens the black box of absolute depth estimation, which is convenient for analyzing the exceptions in the application.

V. EXPERIMENTS

A. Datasets and Evaluation Metrics

This paper evaluates two datasets most commonly used in the 3D human pose estimation tasks: Human 3.6M [35] and MPI-INF-3DHP [36].

1) *Human 3.6M*: It is the most classical and widely used indoor dataset. It uses four 50 Hz cameras to capture a total of 3.6 million frames and collects the 2D ground truth pose and 3D ground truth pose of the human body corresponding to each frame of images. Following previous works [11], [13], [14], this paper train the proposed model on subjects S1, S5, S6, S7, and S8 and test on subjects S9 and S11. This work also used performance metrics from previous work on the Human 3.6M: the Mean Per Joint Position Error (MPJPE) and the MPJPE after aligning predictions with ground truth 3D pose by a similarity transformation (P-MPJPE).

2) *MPI-INF-3DHP*: It is a large dataset containing indoor and outdoor data, which increases the challenge of the test task. The training set contains eight activities of 8 subjects, and the test set contains seven activities in the three scenes green screen, non-green screen, and outdoor. Following previous works [10], [14], this paper uses all activities from 8 camera views in the train set for training and testing the whole test set. Besides MPJPE metrics, two other metrics are often applied to 3DHP datasets: the Percentage of Correct Keypoints (PCK) within the 150mm range and the Area Under Curve (AUC).

B. Statistical Characteristics of Datasets

This paper statistics the probability of depth ambiguity on two real datasets, Human 3.6M and MPI-INF-3DHP, to reveal the statistical characteristics of symbol singularity in Equation 15. Specifically, our work linearly projects the 3D poses from two datasets onto the imaging plane using the given camera's internal parameters to calculate the ground truth 2D pose. Given the 3D and 2D poses, the depth information from all non-root nodes to root nodes can be calculated using Equations 14, 15, and 16, including whether the appearance of depth ambiguity and the selection of symbols in Equation 15 after ambiguity occurs.

In the Human 3.6M, this paper statistics 58.6 million corresponding depth data between the non-root joint and root joint. Among them, 0.17 million pieces of data may occur depth ambiguity, accounting for 0.29%; 0.05 million pieces of data were selected with a negative sign in Equation 15 during depth calculation, accounting for 0.08%; 36.66 million pieces of data with acute angle α , accounting for 62.56%; in the case of acute angle α , the probability of AB is less than BE , accounting for 0.46%, which may be the reason for the low possibility of depth ambiguity occurrence.

Similarly, this paper also statistics 34.08 million corresponding depth data between the non-root joint and to root joint in MPI-INF-3DHP. Among them, 1.38 million pieces of data may occur depth ambiguity, accounting for 4.06%; 0.36 million pieces of data were selected with a negative sign

in Equation 15 during depth calculation, accounting for 1.06%; 20.02 million pieces of data with acute angle α , accounting for 58.73%; in the case of acute angle α , the probability of AB is less than BE , accounting for 6.91%.

These statistical data reveal the reason for the low proportion of deep ambiguity in the samples, that is, the geometric feature AE in real data is usually greater than BE . These evidences also support our hypothesis that the probability of taking symbols in Equation 15 is low.

C. Model Implementation Details

The model hyperparameters are shown in Section IV-A.1. The trimmed coefficient k is set to 4 in the absolute depth estimation algorithm. The proposed model is trained on one Tesla V100 GPU. An AdamW optimizer is used to optimize the model. Its initial learning rate is set to 0.001, and the decay rate of each epoch is 0.99. The training process has been iterated for 30 epochs. Following previous works [10], [13], [14], this paper adopted horizontal flip augmentation during testing.

D. Comparison With State-of-the-Art Methods

1) *Result on Human 3.6M (H36M) Dataset*: The MPJPE results with the 2D pose detected by CPN and the ground truth 2D pose are reported in Table I. The proposed model achieves the state-of-the-art result under the MPJPE (34.0 mm) using ground truth 2D pose. Furthermore, its accuracy is second only to the newly proposed JointFormer [13] (50.4 mm vs. 50.1 mm) by inputting the detected 2D pose (CPN). It is worth noting that this result does not employ the refinement technique similar to Lutz et al. [13] or Cai et al. [15]. Such refinement designs spend additional parameters and only improve the results by a few tenths of a millimeter. This paper reports P-MPJPE results with the 2D pose detected by CPN in Table II. The proposed model achieves excellent performance (39.4 mm in P-MPJPE), outperforming lots of traditional works.

In addition, this paper compares the proposed non-learning-based method with the traditional learning-based methods under the absolute depth estimation task on the Human 3.6M. The root absolute MPJPE and the non-root MPJPE results are reported in Table III. The proposed non-learning-based algorithm achieves a similar accuracy of absolute depth estimation as the learning-based method in most cases. It can be perfectly embedded into existing single-frame or video-based pose estimation models without any parameters, showing powerful universality and portability. Except for individual cases, as the precision of relative pose estimation increases, the depth estimation error tends to decrease, consistent with the theoretical analysis. It is emphasized that the traditional simultaneous prediction mode will decline relative 3D pose estimation accuracy by 1 to 2 mm. Nevertheless, the proposed algorithm will not have any impact.

Figure 4 shows visualization results on the Human 3.6M. Although limited by single frame input, the proposed model performs well under slight occlusion. Another challenge of pose estimation occurs in bending or sitting. When bending,

TABLE I

QUANTITATIVE COMPARISON WITH THE STATE-OF-THE-ART METHOD ON HUMAN 3.6M UNDER MPJPE (MM) USING GROUND TRUTH 2D POSE (GT) OR DETECTED 2D POSE (CPN) AS INPUT. **BLOD**: BEST; UNDERLINED: SECOND BEST

Method	Dir.	Disc.	Eat	Greet	Phone	Photo	Pose	Purch.	Sit	SitD.	Smoke	Wait	WalkD.	Walk	WalkT.	Avg.
Martinez. et al. (ICCV17)[5]	37.7	44.4	40.3	42.1	48.2	54.9	44.4	42.1	54.6	58.0	45.1	46.4	47.6	36.4	40.4	45.5
Hossain et al. (ECCV18)[37]	35.2	40.8	37.2	37.4	43.2	44.0	38.9	35.6	42.3	44.6	39.7	39.7	40.2	32.8	35.5	39.2
Zhao et al. (CVPR19)[26]	37.8	49.4	37.6	40.9	45.1	41.4	40.1	48.3	50.1	42.2	53.5	44.3	40.5	47.3	39.0	43.8
Ci et al. (ICCV19)[38]	36.3	38.8	29.7	37.8	<u>34.6</u>	42.5	39.8	32.5	36.2	39.5	34.4	38.4	38.2	31.3	34.2	36.3
Liu et al. (ECCV20)[39]	36.8	40.3	33.0	36.3	37.5	45.0	39.7	34.9	40.3	47.7	37.4	38.5	38.6	29.6	32.0	37.8
Xu et al. (CVPR21)[11]	35.8	38.1	31.0	35.3	35.8	43.2	37.3	31.7	38.4	45.5	35.4	36.7	36.8	27.9	30.7	35.8
Zhao et al. (CVPR22)[14]	32.0	38.0	30.4	34.4	34.7	43.3	35.2	31.4	38.0	46.2	34.2	35.7	36.1	27.4	30.6	<u>35.2</u>
Lutz et al. (ICPR22)[13]	31.0	36.6	30.2	33.4	33.5	39.0	37.1	31.3	<u>37.1</u>	<u>40.1</u>	<u>33.8</u>	33.5	<u>35.0</u>	28.7	29.1	34.0
Our (GT)	30.0	35.5	<u>30.1</u>	31.9	34.8	37.3	35.4	31.7	39.5	42.9	33.3	<u>34.4</u>	34.6	28.5	29.9	34.0
Martinez. et al. (ICCV17)[5]	51.8	56.2	58.1	59.0	69.5	78.4	55.2	58.1	74.0	94.6	62.3	59.1	65.1	49.5	52.4	62.9
Pavlakos et al. (ECCV18)[22]	48.5	54.4	54.5	52.0	59.4	65.3	49.9	52.9	65.8	71.1	56.6	52.9	60.9	44.7	47.8	56.2
Sharma et al. (ICCV19)[40]	48.6	54.5	54.2	55.7	62.2	72.0	50.5	54.3	70.0	78.3	58.1	55.4	61.4	45.2	49.7	58.0
Ci et al. (ICCV19)[38]	46.8	52.3	44.7	50.4	52.9	68.9	49.6	46.4	60.2	78.9	51.2	50.0	54.8	40.4	43.3	52.7
Liu et al. (ECCV20)[39]	46.3	52.2	47.3	50.7	55.5	67.1	49.2	46.0	60.4	71.1	51.5	50.1	54.5	40.3	43.7	52.4
Xu et al. (CVPR21)[11]	45.2	<u>49.9</u>	47.5	50.9	54.9	66.1	48.5	46.3	59.7	71.5	51.4	48.6	53.9	39.9	44.1	51.9
Zhao et al. (CVPR22)[14]	45.2	50.8	48.0	50.0	54.9	65.0	<u>48.2</u>	47.1	60.2	70.0	51.6	48.7	54.1	39.7	43.1	51.8
Lutz et al. (ICPR22)[13]	45.6	49.7	<u>46.0</u>	49.3	<u>52.2</u>	<u>58.8</u>	47.5	<u>46.1</u>	58.2	66.1	50.7	47.5	52.6	39.2	41.6	50.1
Our (CPN)	<u>45.3</u>	51.8	46.4	<u>49.9</u>	51.4	57.6	49.1	46.5	<u>59.1</u>	<u>67.9</u>	<u>51.0</u>	48.9	<u>53.0</u>	38.0	40.6	<u>50.4</u>

TABLE II

QUANTITATIVE COMPARISON WITH THE STATE-OF-THE-ART METHOD ON HUMAN 3.6M UNDER P-MPJPE (MM) USING DETECTED 2D POSE (CPN) AS INPUT. **BLOD**: BEST; UNDERLINED: SECOND BEST

Method	Dir.	Disc.	Eat	Greet	Phone	Photo	Pose	Purch.	Sit	SitD.	Smoke	Wait	WalkD.	Walk	WalkT.	Avg.
Martinez et al. (ICCV17) [5]	39.5	43.2	46.4	47.0	51.0	56.0	41.4	40.6	56.5	69.4	49.2	45.0	49.5	38.0	43.1	47.7
Sun et al. (ICCV17)[41]	42.1	44.3	45.0	45.4	51.5	53.0	43.2	41.3	59.3	73.3	51.0	44.0	48.0	38.3	44.8	48.3
Fang et al. (AAAI18)[42]	38.2	41.7	43.7	44.9	48.5	55.3	40.2	38.2	<u>45.5</u>	64.4	47.2	44.3	47.3	36.7	41.7	45.7
Pavlakos et al. (ECCV18)[22]	34.7	39.8	41.8	38.6	42.5	47.5	38.0	36.6	50.7	56.8	42.6	39.6	43.9	32.1	36.5	41.8
Li et al. (CVPR19)[43]	35.5	39.8	41.3	42.3	46.0	48.9	36.9	37.3	51.0	60.6	44.9	40.2	44.1	33.1	36.9	42.6
Ci et al. (ICCV19)[38]	36.9	41.6	38.0	41.0	41.9	51.1	38.2	37.6	49.1	62.1	43.1	39.9	43.5	32.2	37.0	42.2
Cai et al. (ICCV19)[15]	36.8	<u>38.7</u>	38.2	41.7	40.7	46.8	37.9	35.6	47.6	51.7	41.3	36.8	42.7	31.0	34.7	40.2
Pavlo et al. (CVPR19)[31]	36.0	<u>38.7</u>	38.0	41.7	40.1	45.9	37.1	<u>35.4</u>	46.8	53.4	41.4	36.9	43.1	30.3	34.8	40.0
Liu et al. (ECCV20)[39]	35.9	40.0	38.0	41.5	42.5	51.4	37.8	<u>36.0</u>	48.6	56.6	41.8	38.3	42.7	31.7	36.2	41.2
Lutz et al. (ICPR22)[13]	34.9	38.5	35.3	39.9	39.8	44.3	36.1	35.1	43.3	52.0	40.2	36.1	41.9	30.3	33.8	39.0
Our	35.4	38.8	<u>35.9</u>	40.4	39.6	44.0	<u>36.7</u>	<u>35.4</u>	46.8	53.7	40.9	<u>36.6</u>	42.0	<u>30.6</u>	<u>33.9</u>	39.4

TABLE III

QUANTITATIVE COMPARISON WITH THE PROPOSED NON-LEARNING-BASED AND TRADITIONAL LEARNING-BASED METHODS ON HUMAN 3.6M UNDER THE ABS-MPJPE(MM) AND THE MPJPE(MM)

Method	Detected 2D Pose (CPN)				Ground Truth 2D Pose			
	Learning-based		Non-learning-based (our)		Learning-based		Non-learning-based (our)	
	Root Abs-MPJPE	Non-root MPJPE	Root Abs-MPJPE	Non-root MPJPE	Root Abs-MPJPE	Non-root MPJPE	Root Abs-MPJPE	Non-root MPJPE
Li et al. (CVPR22) (f=27)[9]	122.2	47.3	127.6	46.0	90.3	35.7	98.1	34.6
Pavlo et al. (CVPR19) (f=1)[31]	142.2	55.0	142.8	54.3	117.7	41.6	101.0	40.4
Zheng et al. (ICCV21) (f=1) [28]	134.5	52.1	133.5	51.8	94.1	40.1	98.6	38.4
Lutz et al. (ICPR22) (f=1) [13]	131.3	52.2	130.2	50.1	98.1	36.2	116.3	34.0
Our	-	-	134.0	50.4	-	-	96.1	34.0

the 2D key points of the face and the 2D key points of the torso are located densely. Under such cases, the proposed model still has excellent robustness to restore the upright state of the human body with high quality. Similarly, the proposed model can correctly estimate the 3D coordinates of entangled feet in sitting.

2) *Result on MPI-INF-3DHP (3DHP) Dataset*: This paper compares the proposed model with existing state-of-the-art methods on MPI-INF-3DHP. On the one hand, this paper trains the model on the Human 3.6M and tests it on the MPI-INF-3DHP to evaluate the model's generalization ability across scenarios. The experimental results are reported in the upper part of Table IV. The proposed method outperforms all previous state-of-the-art methods under PCK (85.0 vs. 83.7) and AUC (53.1 vs. 52.4) and achieves MPJPE (84.1 mm), which proves the solid cross-scene generalization ability. On the other hand,

many recent works based on video sequences have applied the training dataset in MPI-INF-3DHP for training. This paper also compares these models and reports the results in the lower part of Table IV. The proposed model achieved excellent accuracy in three evaluation metrics: PCK (96.7), AUC (75.6), and MPJPE (38.2 mm). It means that the proposed model is more competitive than many video-based methods on the MPI-INF-3DHP benchmark. Figure 5 shows the visualization results of the proposed model and the state-of-the-art method (JointFormer [13]) on the challenging pose under the MPI-INF-3DHP.

E. Ablation Study

1) *Discussion on Model Parameters*: This paper reports the results with different network layer numbers in the top part of Table V. Experiments show that increasing the network

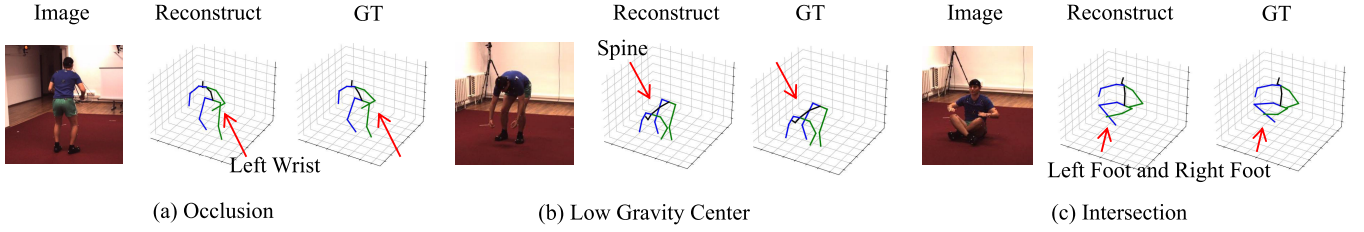


Fig. 4. Visualize the reconstruction results of our model on Human 3.6M. (a) The occluded left wrist. (b) The straight spine when bending. (c) The leg intersects when sitting.

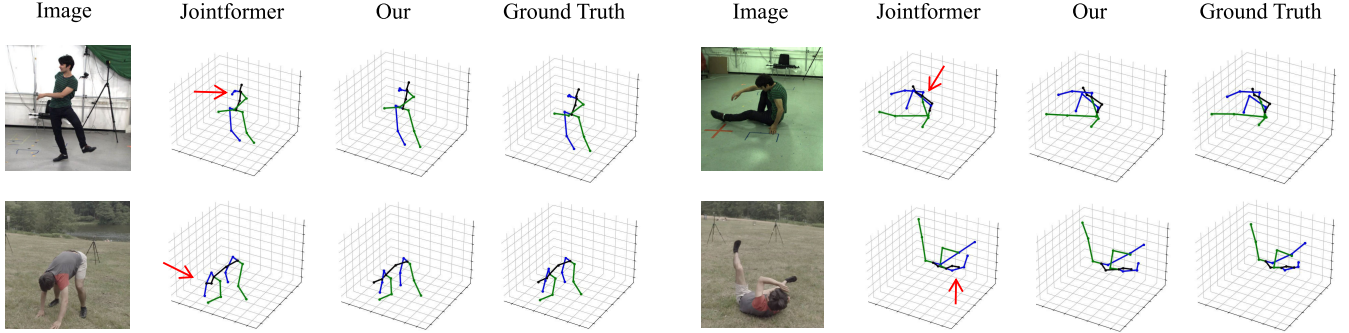


Fig. 5. Qualitative comparison with the proposed method and the previous state-of-the-art method (JointFormer [13]) on MPI-INF-3DHP. Wrong estimations are highlighted by arrows.

TABLE IV

QUANTITATIVE COMPARISON WITH THE STATE-OF-THE-ART METHOD ON MPI-INF-3DHP UNDER PCK, AUC, AND MPJPE (MM). **BLOD**: BEST; UNDERLINED: SECOND BEST. F DENOTES THE FRAME NUMBER

Train Dataset	Method	PCK	AUC	MPJPE
H36M	Mehta et al. (3DV17) [36]	64.7	31.7	-
	Luo et al. (BMVC18) [44]	65.6	33.2	-
	Ci et al. (ICCV19) [38]	74.0	36.7	-
	Xu et al. (CVPR21) [11]	80.1	45.8	-
	Zhao et al. (CVPR22) [14]	79.0	43.8	-
	Lutz et al. (ICPR22) [13]	83.7	52.4	-
	Our	85.0	53.1	84.1
3DHP	Zeng et al. (f=1)(ECCV20) [45]	77.6	43.8	-
	Wang et al. (f=96)(ECCV20) [46]	86.9	62.1	68.1
	Zheng et al. (f=9)(ICCV21) [28]	88.6	62.1	68.1
	Chen et al. (f=81)(TCSVT22) [47]	87.9	54.0	78.8
	Li et al. (f=9)(CVPR22) [9]	93.8	63.3	58.0
	Shan et al. (f=81)(ECCV22) [10]	97.9	75.8	32.2
	Our (f=1)	<u>96.7</u>	<u>75.6</u>	<u>38.2</u>

layers can effectively improve the performance with a slight computation increase. In our motivation, the model should pay more attention to the pre-decoupled geometric features rather than re-learning from the absolute projection coordinates. Therefore, the default parameters design the second's dimensions four times more than the first branches to emphasize the importance of geometric features in Section IV-A.1. To show the effectiveness of such a design more intuitively, this paper tested the impact of different branch dimensions on the results and reported the results in the lower part of Table V. Under similar model sizes, giving more hidden layer parameters to the second branch will improve more obvious precision.

TABLE V

ABLATION STUDY ON DIFFERENT HYPERPARAMETERS OF THE PROPOSED MODEL. EXPERIMENT EVALUATIONS ON HUMAN 3.6M

Layers	First Branch Dimension	Second Branch Dimension	Params (M)	Flops (M)	MPJPE (mm)
2	32	128	0.32	5.53	51.97
4	32	128	0.62	10.74	51.12
6	32	128	0.92	15.94	50.92
8	32	128	1.21	21.01	50.43
8	128	32	1.21	21.01	51.56
8	64	128	1.48	25.69	50.71
8	128	64	1.48	25.69	51.01
8	128	128	2.41	41.74	50.52

TABLE VI

ABLATION STUDY ON DIFFERENT COMPONENTS. EXPERIMENT EVALUATIONS ON HUMAN 3.6M

Exp.	First Branch	Second Branch	Output Joint Vector	Output 3D Pose	MPJPE (mm)
Exp. 1	✓	×	×	✓	52.94
Exp. 2	✓	×	✓	×	52.25
Exp. 3	✓	✓	×	✓	50.91
Original	✓	✓	✓	×	50.43

2) *Effects of Architecture*: This paper tested the effectiveness of each component of the proposed method. For comparison, the first two controls group remove the second branch, and the 3D pose and joint vectors are predicted, respectively. Moreover, the third control group retains both branches but only predicts a single pose. Both experiments use similar model sizes. As shown in Table VI, the proposed method reduces the MPJPE of the three groups of experiments by 2.51, 1.82, and 0.48, respectively. It proves that the explicit

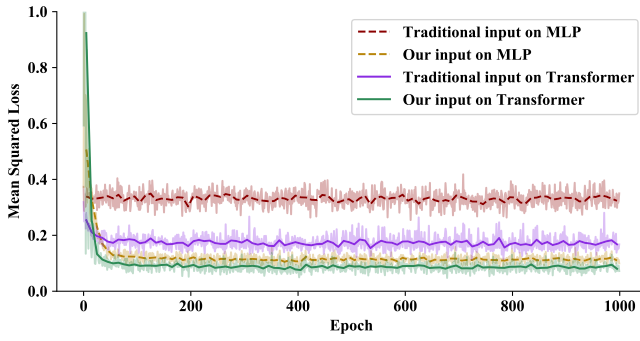


Fig. 6. Results of relative depth estimation experiment with different inputs. The curve with low background transparency represents the actual loss of each iteration. The foreground curve represents the moving average loss with a step of 5.

TABLE VII

ABLATION STUDY ON DIFFERENT GEOMETRIC FEATURE INPUTS. EXPERIMENT EVALUATIONS ON HUMAN 3.6M

Exp.	OD	CD	$\cos \alpha$	MPJPE (mm)
Exp. 1	✓	×	×	51.31
Exp. 2	×	✓	×	51.40
Exp. 3	×	×	✓	51.09
Exp. 4	✓	✓	×	51.13
Exp. 5	✓	×	✓	50.77
Exp. 6	×	✓	✓	50.94
Original	✓	✓	✓	50.43

utilization of geometric features can significantly improve the accuracy, and generating multiple feasible solutions using 3D joint vectors can slightly reduce the error.

Theoretically, neural networks have a solid non-linear fitting ability to learn these non-linear geometric features. However, as Section III mentioned, multiple optimization routes may exist to complete the 2D-to-3D lifting task. The model may not find a decoupled geometric-based optimization route without guidance. Therefore, this paper further designs a simple experiment to prove that the explicit use of geometric information is better than the traditional method using absolute coordinates as input.

This work randomly samples a unit vector in three-dimensional space to simulate human bones and project the two endpoints into a two-dimensional plane. The difference between the depth values of two endpoints in 3D space is set as a target, and two inputs are designed, respectively. The first only takes the absolute coordinates of two endpoints on the plane as input, and the other is the geometric feature as input which is mentioned in Section IV-A.2. The two sets of inputs use the same model architecture and hidden layer dimensions. The optimization results are reported in Figure 6. The proposed method is generally higher than the traditional input in multi-layer perceptron (MLP) and Transformer models. This means that when there are few parameters and a lack of guidance, the neural network is difficult to decouple the depth relationship between joints. It further proves the effectiveness of this work.

3) *Effects of Non-Linear Geometric Features*: This paper tested the effect of different geometric feature inputs on the results to discover the importance of each proposed geometric

TABLE VIII

ABLATION STUDY ON DIFFERENT TRIMMED COEFFICIENT K . EXPERIMENT EVALUATIONS ON HUMAN 3.6M

Backbone	k=1	k=2	k=3	k=4	k=5	k=6
Our (CPN)	166.9	138.3	134.2	134.0	134.5	135.0
Our (GT)	113.0	100.6	96.1	96.1	96.3	96.5
JointFormer (CPN)	182.6	135.2	130.5	130.2	130.3	130.5
JointFormer (GT)	141.9	124.4	116.7	116.3	116.5	116.9

feature. We mainly focus on the non-linear characteristics of the last three dimensions because the features of the first two dimensions are commonly utilized in graph networks and are not the main contributions of this paper. The results of different geometric feature inputs are reported in Table VII. The experiment shows that all proposed geometric features are beneficial to improve the model's accuracy. When the geometric features of OD , CD , and $\cos \alpha$ are deleted independently, the model accuracy decreases by 0.51, 0.34, and 0.7, respectively. It can be considered that $\cos \alpha$ has the most significant impact on the proposed model results, followed by OD . As the theoretical analysis shows, the sign of the former can be used to determine the relative depth between two nodes, and these two features can be used simultaneously to determine the conditions for the occurrence of depth ambiguity.

4) *Ablation Study on the Root Node Absolute Depth Estimation Algorithm*: This paper tested the effect of different trimmed coefficients k on the root node absolute depth estimation results. As shown in Table VIII, bigger or smaller trimmed coefficients k led to increased algorithm error. Theoretically, many outliers will be retained in the depth candidate list when k tends to be smaller. And the algorithm will highly rely on the depth estimation of individual joints when k tends to be larger. The former reduces the model's accuracy, and the latter reduces the robustness of the model. The experimental results are consistent with the theoretical analysis. According to the experience of employing the algorithm on different backbone models, the initial trimmed coefficient k is set to 4.

Another factor that affects the proposed algorithm is the accuracy of the relative 3D pose estimation, according to Equations 14 and 15. Theoretically, the error of root node absolute depth estimation decreases as the error of relative pose estimation decreases. As shown in Table III and Table VII, after converting from detected 2D poses (based on CPN) to ground truth 2D poses as input, the lower relative pose error helps to reduce the absolute depth error accordingly. At the same time, this paper notes some bad samples. For example, the proposed and state-of-the-art model [13] have the same relative pose estimation accuracy (34 mm in MPJPE) using ground truth 2D pose. Still, the absolute depth of the root node is different (96.1 mm vs. 116.3 mm in absolute MPJPE). The error distribution of the different models is the key to explaining this phenomenon. This paper shows the error histogram of the proposed model and JointFormer [13] under the ground truth 2D poses in Figure 7. In the upper part of Figure 7 (a), the proposed model has the same relative pose estimation error as the JointFormer, so the area of the two distributions is approximate. However, only parts of the joints are utilized in the final root node absolute depth estimation calculation because the trimmed mean method filters out the

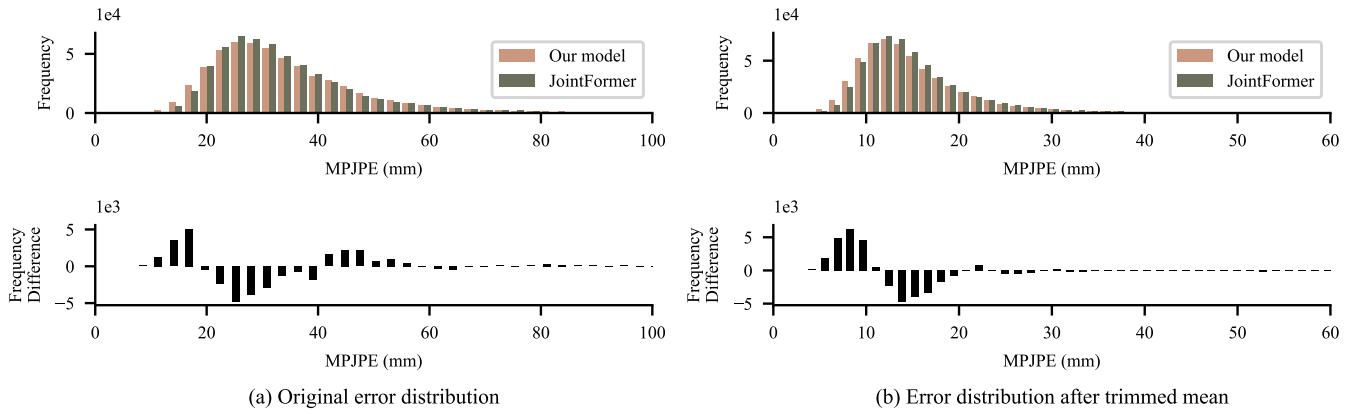


Fig. 7. The error histogram of the proposed model and JointFormer [13] on H36M. (a) The upper figure denotes the original error distribution, and the lower one denotes the difference in frequency between the two models within the error interval. When the frequency difference is positive, it indicates that the sample size of our model in this error interval is greater than the JointFormer, and vice versa. (b) The upper figure denotes the error distribution after trimmed mean, and the lower one denotes the corresponding frequency difference.

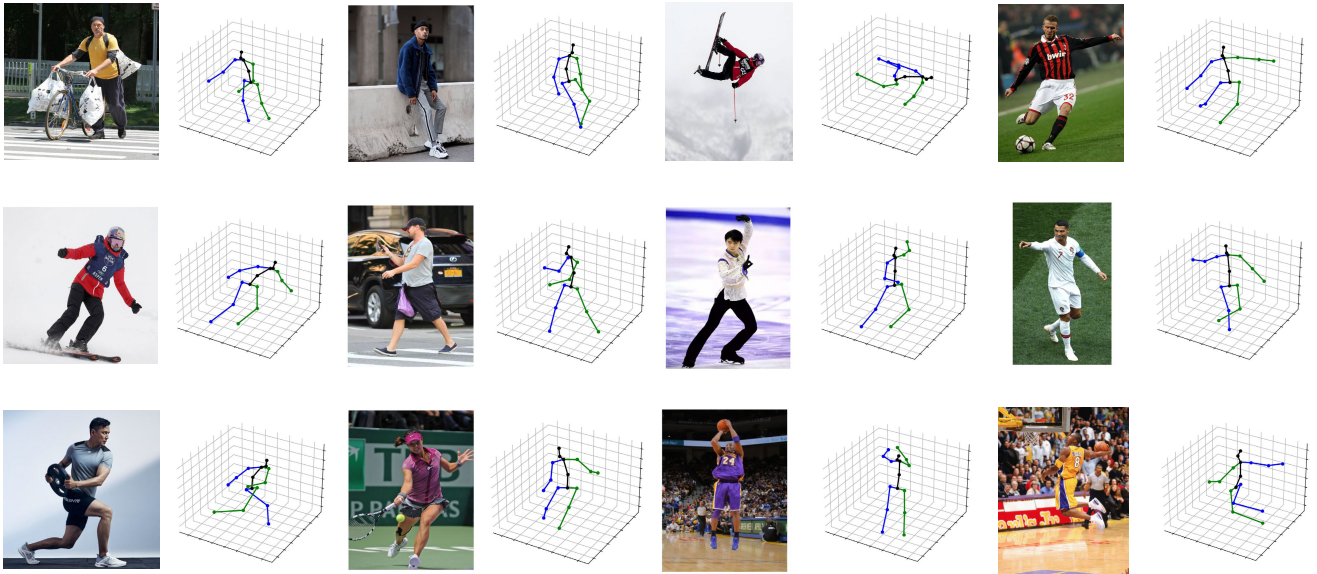


Fig. 8. The visualization results on challenging internet images, including daily behavior and sport.

joints with significant errors. We further explore the error distribution (the upper part) after the trimmed mean and its difference (the lower part) in Figure 7 (b). It can be intuitively observed that the sample size of our method in the interval with significant error after being trimmed mean is smaller than that of Jointformer. That is, it tends to be minor that the error of depth candidate values used in the proposed model's final calculation. It explains the lower error of the proposed model in the root node absolute depth estimation task.

F. Model Size and Running Time

Benefiting from the decoupling of geometric characteristics, the number of the proposed model parameters is less than many existing works, which ensures a faster inference speed. To highlight the computational efficiency of the proposed method, this paper tested the inference speed on a computational-resources-limited platform: Intel i7-10750 CPU. Table IX compared several recent models' performance in model size, inference time, and accuracy.

Single and parallel speeds refer to the average single frame inferring speed when the batch size is set to 1 and 128, respectively. Experiments have confirmed that the proposed model has achieved higher accuracy at a faster processing speed. In particular, when the batch size is 128, the calculation speed of our model reaches 1064 frames per second, which meets the demand for speed in practical applications.

G. More Visualizations Result and Model Application

Due to the limitation of equipment, it is difficult to collect outdoor 3D human poses, which challenges the generalization ability of the pose estimation model. Figure 8 shows visualization results of some images from the internet. The proposed model has performed well in complex outdoor environments, including challenges such as jumping, tilting, and occlusion. It demonstrates that the proposed model has an excellent generalization ability.

We actually deployed the proposed model in a factory to ensure the security of employees. Figure 9 shows some

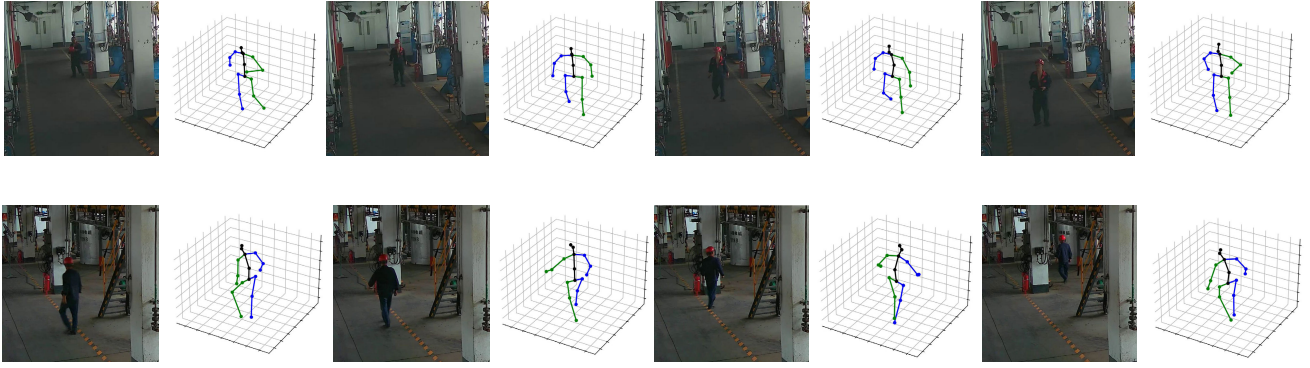


Fig. 9. The visualization results in the factory scenario. The proposed model can reconstruct acceptable 3D poses while maintaining high frame rate detection.

TABLE IX

A COMPARISON OF THE RECENT MODEL'S PARAMETERS, SPEED (FRAMES PER SECOND), AND ACCURACY. $\dagger$ DENOTES THE H36M DATASET AND $\ddagger$ DENOTES THE 3DHP DATASET

Method	Params (M)	Single Speed	Parallel Speed	MPJPE $\dagger$ ↓ (mm)	PCK $\ddagger$ ↑
Xu et al. [11]	3.70	133	761	51.9	80.1
Zhao et al. [14]	1.39	141	1396	51.8	79.0
Lutz et al. [13]	3.82	137	757	50.1	83.7
Our	1.21	125	1064	50.4	85.0

visualization results in the application. The main challenge is that the employees' uniforms tend to be the same color when the light is dark. It will lead to deviation of 2D pose estimation and further affect 3D pose estimation. The prediction result of the proposed model is the integration of multiple feasible solutions, which is more stable than single pose estimation. In addition, benefiting from the proposed model's low parameters and high inferring speed, it maintains a high detection frame rate in the application.

VI. CONCLUSION

This paper proposes a novel perspective for estimating the 3D human pose. In the pinhole imaging system, this work deduced the relative depth expression between two nodes and quantitatively analyzed depth ambiguity's cause and influence factors. The reasonable utilization of nonlinear geometric features based on the variables in the depth expression and the influence factors of depth ambiguity is emphasized in the proposed model. These features and the 2D pose coordinates are fed into a two-branch transformer network, which is light and efficient. In addition, this paper first presents a geometric-based and non-learning-based absolute depth estimation algorithm. It is parameter-free, plug-and-play, and interpretable. Experiments on two benchmarks have confirmed that the proposed pose estimation model can maintain higher accuracy and cross-scene generalization ability with smaller parameters and faster inferring speed than current the-state-of-art models. The ablation study further proves the importance of utilizing geometric features directly in the proposed pose estimation model and analyzes the operation principle of the proposed absolute depth estimation algorithm. The experimental results are consistent with the theoretical analysis. Finally, this paper makes a qualitative analysis of more challenging

unstructured datasets. The proposed model can efficiently reconstruct convincing 3D pose results in occluded, outdoor, and ambiguous scenes, proving its strong generalization ability and high practical significance.

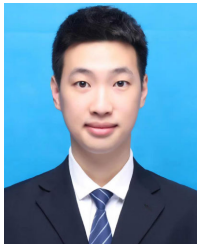
Consistent with lots of single-frame 2D-to-3D lifting pose estimation frameworks, the main limitation of the current model is its sensitivity to input 2D pose accuracy. The single-frame 2D pose lacks image semantic information and temporal consistency from context, which are open problems. The proposed method focuses on prior geometric knowledge to decouple skeleton spatial features, which is compatible with current image-processing and video-based temporal methods. In applications with sufficient computing power and non-instantaneous detection, subsequent work may extend our work with effective image-processing or temporal decoupling methods to alleviate the above limitations.

In addition, there are also two problems that can be further studied by follow-up work. First, the follow-up work can further understand and alleviate the depth of ambiguity based on geometric relations. In Equation 8, this paper points out the conditions for the ambiguity of 2D-to-3D mapping and that this condition is harsh. Therefore, these ambiguous samples have a small proportion in the training sets. Thus, the pose estimation model has insufficient ambiguity learning for 2D-to-3D mapping. Follow-up work can consider how to alleviate the precision loss caused by depth ambiguity from the perspective of the hard sample learning method. Secondly, the follow-up work can study a completely non-learning-based pose estimation model based on Equation 9. Equation 9 proves that given the absolute depth of a joint, the relative joint coordinates of the joint connected to it can be explicitly calculated. The absolute depth of the joint can be obtained by the RGBD camera. This method requires researchers to consider further the technique of overcoming depth ambiguity. This paper mainly provides a geometry-based viewpoint on the 3D human pose estimation task. Subsequent work can also employ more powerful models combined with geometric ideas to improve the model's accuracy.

REFERENCES

- [1] X. Zhang, M. Ding, and G. Fan, "Video-based human walking estimation using joint gait and pose manifolds," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 27, no. 7, pp. 1540–1554, Jul. 2017.

- [2] M. Li, S. Chen, X. Chen, Y. Zhang, Y. Wang, and Q. Tian, "Actional-structural graph convolutional networks for skeleton-based action recognition," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 3595–3603.
- [3] J. Gan and W. Wang, "In-air handwritten English word recognition using attention recurrent translator," *Neural Comput. Appl.*, vol. 31, no. 7, pp. 3155–3172, Jul. 2019.
- [4] L. Dong, X. Chen, R. Wang, Q. Zhang, and E. Izquierdo, "ADORE: An adaptive holons representation framework for human pose estimation," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 28, no. 10, pp. 2803–2813, Oct. 2018.
- [5] J. Martinez, R. Hossain, J. Romero, and J. J. Little, "A simple yet effective baseline for 3D human pose estimation," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2659–2668.
- [6] W. Li, H. Liu, R. Ding, M. Liu, P. Wang, and W. Yang, "Exploiting temporal contexts with strided transformer for 3D human pose estimation," *IEEE Trans. Multimedia*, vol. 25, pp. 1282–1293, 2023.
- [7] S. Yan, Y. Xiong, and D. Lin, "Spatial temporal graph convolutional networks for skeleton-based action recognition," in *Proc. AAAI Conf. Artif. Intell.*, 2018, pp. 7444–7452.
- [8] M. Kocabas, N. Athanasiou, and M. J. Black, "VIBE: Video inference for human body pose and shape estimation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 5253–5263.
- [9] W. Li, H. Liu, H. Tang, P. Wang, and L. Van Gool, "MHFormer: Multi-hypothesis transformer for 3D human pose estimation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 13147–13156.
- [10] W. Shan, Z. Liu, X. Zhang, S. Wang, S. Ma, and W. Gao, "P-STMO: Pre-trained spatial temporal many-to-one model for 3D human pose estimation," in *Proc. Conf. Eur. Conf. Comput. Vis. (ECCV)*, 2022, pp. 461–478.
- [11] T. Xu and W. Takano, "Graph stacked hourglass networks for 3D human pose estimation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 16105–16114.
- [12] Z. Zou and W. Tang, "Modulated graph convolutional network for 3D human pose estimation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 11477–11487.
- [13] S. Lutz, R. Blythman, K. Ghosal, M. Moynihan, C. Simms, and A. Smolic, "JointFormer: Single-frame lifting transformer with error prediction and refinement for 3D human pose estimation," in *Proc. 26th Int. Conf. Pattern Recognit. (ICPR)*, Aug. 2022, pp. 1–8.
- [14] W. Zhao, W. Wang, and Y. Tian, "GraFormer: Graph-oriented transformer for 3D pose estimation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 20406–20415.
- [15] Y. Cai et al., "Exploiting spatial-temporal relationships for 3D pose estimation via graph convolutional networks," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 2272–2281.
- [16] C. Sminchisescu, "3D human motion analysis in monocular video techniques and challenges," in *Proc. IEEE Int. Conf. Video Signal Based Surveill.*, 2006, p. 76, doi: [10.1109/AVSS.2006.3](https://doi.org/10.1109/AVSS.2006.3).
- [17] A. Agarwal and B. Triggs, "Recovering 3D human pose from monocular images," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 28, no. 1, pp. 44–58, Jan. 2006.
- [18] G. Wei, C. Lan, W. Zeng, and Z. Chen, "View invariant 3D human pose estimation," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 30, no. 12, pp. 4601–4610, Dec. 2020.
- [19] S. Li and A. B. Chan, "3D human pose estimation from monocular images with deep convolutional neural network," in *Proc. Asian Conf. Comput. Vis. (ACCV)*, 2015, pp. 332–347.
- [20] X. Zheng, X. Chen, and X. Lu, "A joint relationship aware neural network for single-image 3D human pose estimation," *IEEE Trans. Image Process.*, vol. 29, pp. 4747–4758, 2020.
- [21] S. Sedai, M. Bennamoun, and D. Q. Huynh, "A Gaussian process guided particle filter for tracking 3D human pose in video," *IEEE Trans. Image Process.*, vol. 22, no. 11, pp. 4286–4300, Nov. 2013.
- [22] G. Pavlakos, X. Zhou, and K. Daniilidis, "Ordinal depth supervision for 3D human pose estimation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7307–7316.
- [23] W. Yang, W. Ouyang, X. Wang, J. Ren, H. Li, and X. Wang, "3D human pose estimation in the wild by adversarial learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 5255–5264.
- [24] R. Gu, G. Wang, Z. Jiang, and J. Hwang, "Multi-person hierarchical 3D pose estimation in natural videos," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 30, no. 11, pp. 4245–4257, Nov. 2020.
- [25] M. F. Ghezalghieh, R. Kasturi, and S. Sarkar, "Learning camera viewpoint using CNN to improve 3D body pose estimation," in *Proc. 4th Int. Conf. 3D Vis. (3DV)*, Oct. 2016, pp. 685–693.
- [26] L. Zhao, X. Peng, Y. Tian, M. Kapadia, and D. N. Metaxas, "Semantic graph convolutional networks for 3D human pose regression," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 3425–3435.
- [27] A. Zeng, X. Sun, L. Yang, N. Zhao, M. Liu, and Q. Xu, "Learning skeletal graph neural networks for hard 3D pose estimation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, p. 11436.
- [28] C. Zheng, S. Zhu, M. Mendieta, T. Yang, C. Chen, and Z. Ding, "3D human pose estimation with spatial and temporal transformers," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, p. 11656.
- [29] W. Li, H. Liu, T. Guo, R. Ding, and H. Tang, "GraphMLP: A graph MLP-like architecture for 3D human pose estimation," 2022, *arXiv:2206.06420*.
- [30] I. Tolstikhin et al., "MLP-mixer: An all-MLP architecture for vision," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 34, 2021, pp. 24261–24272.
- [31] D. Pavlo, C. Feichtenhofer, D. Grangier, and M. Auli, "3D human pose estimation in video with temporal convolutions and semi-supervised training," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 7753–7762.
- [32] M. Véges and A. Lorincz, "Absolute human pose estimation with depth prediction network," in *Proc. Int. Joint Conf. Neural Netw. (IJCNN)*, Jul. 2019, pp. 1–7.
- [33] Y. Zhan, F. Li, R. Weng, and W. Choi, "Ray3D: Ray-based 3D human pose estimation for monocular absolute 3D localization," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 13116–13125.
- [34] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, 2017, pp. 5998–6008.
- [35] C. Ionescu, D. Papava, V. Olaru, and C. Sminchisescu, "Human3.6M: Large scale datasets and predictive methods for 3D human sensing in natural environments," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 36, no. 7, pp. 1325–1339, Jul. 2014.
- [36] D. Mehta et al., "Monocular 3D human pose estimation in the wild using improved CNN supervision," in *Proc. Int. Conf. 3D Vis. (3DV)*, Oct. 2017, pp. 506–516.
- [37] M. R. I. Hossain and J. J. Little, "Exploiting temporal information for 3D human pose estimation," in *Proc. Conf. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 69–86.
- [38] H. Ci, C. Wang, X. Ma, and Y. Wang, "Optimizing network structure for 3D human pose estimation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 2262–2271.
- [39] K. Liu, R. Ding, Z. Zou, L. Wang, and W. Tang, "A comprehensive study of weight sharing in graph networks for 3D human pose estimation," in *Proc. Conf. Eur. Conf. Comput. Vis. (ECCV)*, 2020, pp. 318–334.
- [40] S. Sharma, P. T. Varigonda, P. Bindal, A. Sharma, and A. Jain, "Monocular 3D human pose estimation by generation and ordinal ranking," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 2325–2334.
- [41] X. Sun, J. Shang, S. Liang, and Y. Wei, "Compositional human pose regression," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2621–2630.
- [42] H. Fang, Y. Xu, W. Wang, X. Liu, and S.-C. Zhu, "Learning pose grammar to encode human body configuration for 3D pose estimation," in *Proc. AAAI Conf. Artif. Intell.*, 2018, pp. 6821–6828.
- [43] C. Li and G. H. Lee, "Generating multiple hypotheses for 3D human pose estimation with mixture density network," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 9887–9895.
- [44] C. Luo, X. Chu, and A. Yuille, "OriNet: A fully convolutional network for 3D human pose estimation," in *Proc. Brit. Mach. Vis. Conf. (BMVC)*, 2018, pp. 92–105.
- [45] A. Zeng, X. Sun, F. Huang, M. Liu, Q. Xu, and S. Lin, "SRNet: Improving generalization in 3D human pose estimation with a split-and-recombine approach," in *Proc. Conf. Eur. Conf. Comput. Vis. (ECCV)*, 2020, pp. 507–523.
- [46] J. Wang, S. Yan, Y. Xiong, and D. Lin, "Motion guided 3D pose estimation from videos," in *Proc. Conf. Eur. Conf. Comput. Vis. (ECCV)*, 2020, pp. 764–780.
- [47] T. Chen, C. Fang, X. Shen, Y. Zhu, Z. Chen, and J. Luo, "Anatomy-aware 3D human pose estimation with bone-based pose decomposition," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 32, no. 1, pp. 198–209, Jan. 2022.



Mengxian Hu received the bachelor's degree in mathematics from Changchun Normal University, Changchun, China, in 2019. He is currently pursuing the Ph.D. degree in intelligence science with Tongji University, Shanghai, China. His research interests include deep learning, human pose estimation, and action recognition.



Qingqing Yan (Graduate Student Member, IEEE) received the bachelor's degree in automation engineering from Chang'an University, Xi'an, China, in 2018. She is currently pursuing the Ph.D. degree in control science and engineering with Tongji University, Shanghai, China. Her research interests include deep learning, robot vision, and network compression.



Chengju Liu received the Ph.D. degree in control theory and control engineering from Tongji University, Shanghai, China, in 2011. From October 2011 to July 2012, she was with the BEACON Centre, Michigan State University, East Lansing, MI, USA, as a Research Associate. She is currently a Professor with the College of Electrical and Information Engineering, Tongji University, Shanghai, China. Her research interests include intelligent control, motion control of legged robots, and evolutionary computation.



Qin Fang received the bachelor's degree in automation from the Hefei University of Technology, Hefei, China, in 2019. She is currently pursuing the Ph.D. degree in control science and engineering with Tongji University, Shanghai, China. Her research interests include deep learning, pose estimation, action recognition, and stability control of biped robots.



Shu Li received the bachelor's and master's degrees in control science and engineering from Tongji University, Shanghai, China, in 2015 and 2018, respectively, where he is currently pursuing the Ph.D. degree. From 2018 to 2019, he was an Assistant Engineer with Aviation Industry Corporation of China Ltd. (AVIC), Beijing, China. His research interests include deep learning, robot vision, and network compression.



Qijun Chen (Senior Member, IEEE) received the B.S. degree in automatic control from the Huazhong University of Science and Technology, Wuhan, China, in 1987, the M.S. degree in control engineering from Xi'an Jiaotong University, Xi'an, China, in 1990, and the Ph.D. degree in control theory and control engineering from Tongji University, Shanghai, China, in 1999. He was a Visiting Professor with the University of California at Berkeley, Berkeley, CA, USA, in 2008.